Blind external benchmarks under homology control

A Boltz-2 embedding ΔΔG predictor on S669 and Ssym · three training regimes · MMseqs2 leakage control at 25 % and 30 % identity

S669 is the honest hard test, and most of what looked like a training-corpus effect was leakage or a defective estimator. On S669 with homologues removed, pooled Pearson r is 0.361 / 0.460 / 0.453 for the three regimes, against per-protein medians of 0.59–0.59. On Ssym the apparent FireProt advantage vanishes once shared folds are removed (0.857 ≈ 0.849 ≈ 0.863 on the common-clean subset). The gap between ranking mutations *within* a protein and calibrating *across* proteins is the central quantitative fact these benchmarks expose.

1. Motivation

Held-out splits inside a single corpus measure generalisation to unseen proteins of the same provenance. They cannot measure what happens when the assay, the curation lineage and the protein universe all change at once. S669 and Ssym are the two most widely used blind stability benchmarks, and both are external to this project's training corpora — which makes them the right instrument, provided sequence-identity leakage is controlled rather than assumed away.

Two questions follow. Which training corpus transfers better to an external benchmark, a large set of designed mini-domains or a smaller set of natural proteins? And does the apparent answer survive removing benchmark proteins that are homologous to the training set?

2. Methods

Benchmarks. S669 (541 variants over 62 proteins) is diverse and deliberately dissimilar to common training sets. Ssym (337 variants over 13 proteins) is narrow and dominated by a handful of well-studied folds. Both are capped at 500 residues and every variant's position was validated against the provided sequence. Ssym is direct mutations only; the reverse direction is obtained analytically from the antisymmetry-augmented model.

Regimes. A — trained on Tsuboyama alone (12,359 mutations / 412 proteins). B — trained on FireProt alone (3,205 / 138). D — pretrained on Tsuboyama, warm-start continued on FireProt. Features are the concatenated wild-type and mutant pooled pair-track embeddings; the model is a 5-seed MLP ensemble with antisymmetry augmentation on every training set.

Leakage control. The wild-type sequences of both training corpora and the benchmark are pooled and clustered with MMseqs2 at 25 % and 30 % identity, 80 % coverage. A benchmark protein is leaky with respect to a corpus if it shares a cluster with any protein in it. Three views are reported: *full* (everything), *filtered* (drop proteins homologous to that regime's own training corpus), and *common* (drop proteins homologous to *any* corpus, giving one identical variant subset for all three regimes — the only fair cross-regime comparison).

Sign convention. The benchmarks use the opposite ΔΔG sign; predictions are sign-flipped when the pooled Pearson is negative, and the flip is recorded.

3. Results

3.1 S669 — the hard test

regime	full (541)	filtered (25 %)	common (25 %)
A — Tsuboyama only	0.415 (0.55)	0.415 (0.55)	0.361 (0.59)
B — FireProt only	0.546 (0.59)	0.460 (0.59)	0.460 (0.59)
D — fine-tuned	0.506 (0.54)	0.453 (0.54)	0.453 (0.54)

Pooled Pearson r, with the per-protein median r in parentheses. Regime A has zero overlap with S669, so its filtered column equals its full column. Common-25 holds 360 variants.

Two things separate here. Pooled r — which asks whether the model gets the *magnitude* right across different proteins — sits between 0.36 and 0.46 on the clean subset. The per-protein median, which asks whether it ranks mutations correctly *inside* one protein, sits near 0.59 for every regime. The model orders mutations within a fold considerably better than it places different folds on a common scale.

3.2 Ssym — where the corpus advantage turns out to be leakage

regime	full (337)	filtered (25 %)	common (25 %)
A — Tsuboyama only	0.759 (0.75)	0.759 (0.75)	0.857 (0.75)
B — FireProt only	0.850 (0.85)	0.849 (0.59)	0.849 (0.59)
D — fine-tuned	0.780 (0.76)	0.863 (0.76)	0.863 (0.76)

Pooled Pearson r (per-protein median). Common-25 holds only 47 variants, so it is a weak but unbiased comparison.

On the full set the FireProt-trained regime looks clearly best (0.850 against 0.759). Removing proteins that share a cluster with FireProt collapses that: its per-protein median falls from 0.85 to 0.59, and on the common-clean subset the three regimes are indistinguishable. **The advantage was homology, not training distribution.**

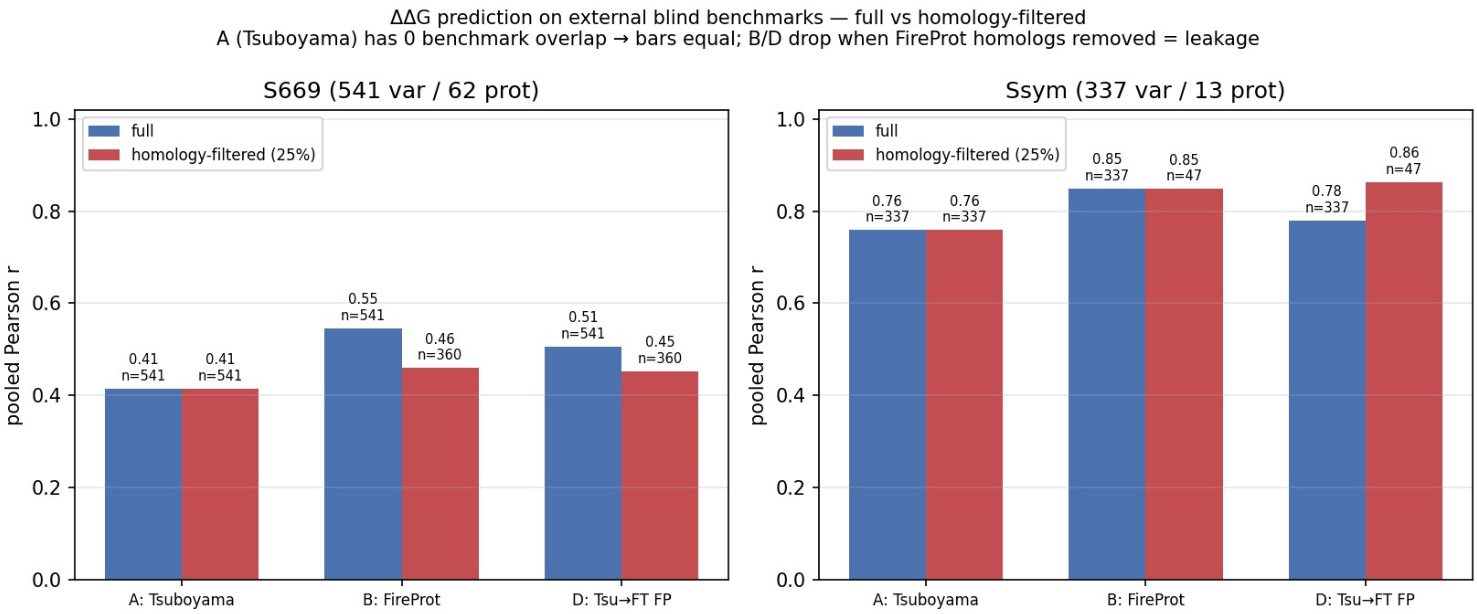

Figure 1. Pooled Pearson r, full versus each regime's own homology filter. Regime A has no benchmark overlap, so its two bars are equal by construction; the drop in B and D is the size of the leakage.

3.3 Antisymmetry: the model is nearly but not exactly antisymmetric

Ssym pairs every forward mutation with its measured reverse, so it tests an identity the model should satisfy: $\Delta\Delta G(A \to B) = -\Delta\Delta G(B \to A)$. Correlation between the direct prediction and the negated reverse is high for all three regimes (0.945 / 0.979 / 0.973), but the residual *bias* is not negligible and it differs by regime: +0.25 kcal/mol for A, +0.04 for B, -0.23 for D. Only FireProt-only training is near-unbiased; training on designed mini-domains alone pushes predictions toward destabilisation, and fine-tuning over-corrects past zero.

3.4 An estimator defect, and what correcting it changed

> The benchmark model was originally fit with early stopping disabled and a fixed 250-iteration budget. Because that budget counts *epochs*, the regime with the most training data took several times as many gradient updates and over-trained hardest — biasing precisely the cross-regime comparison the experiment exists to make. All numbers above use the project's default estimator; the originals are retained for comparison.

benchmark · subset	A Tsuboyama	B FireProt	D fine-tuned
S669 · full	0.255 → **0.415**	0.500 → 0.546	0.462 → 0.506
S669 · common-25	0.214 → **0.361**	0.404 → 0.460	0.408 → 0.453
Ssym · full	0.728 → 0.759	0.891 → 0.850	0.797 → 0.780

Pooled Pearson r before and after the correction. The gain is systematically largest for the regime with the most data, which is the signature the epoch-count argument predicts.

The correction does not change which regime wins, but it halves the size of the corpus effect on S669: the common-25 gap between the FireProt- and Tsuboyama-trained regimes falls from 0.190 to 0.098. It also **reverses one conclusion**: on the defective numbers the fine-tuned regime had the best S669 per-protein median, and on the corrected ones it is last (0.54 against 0.59 and 0.59). Fine-tuning does not earn its keep here — which agrees with the separate within-FireProt result, where it also washed out.

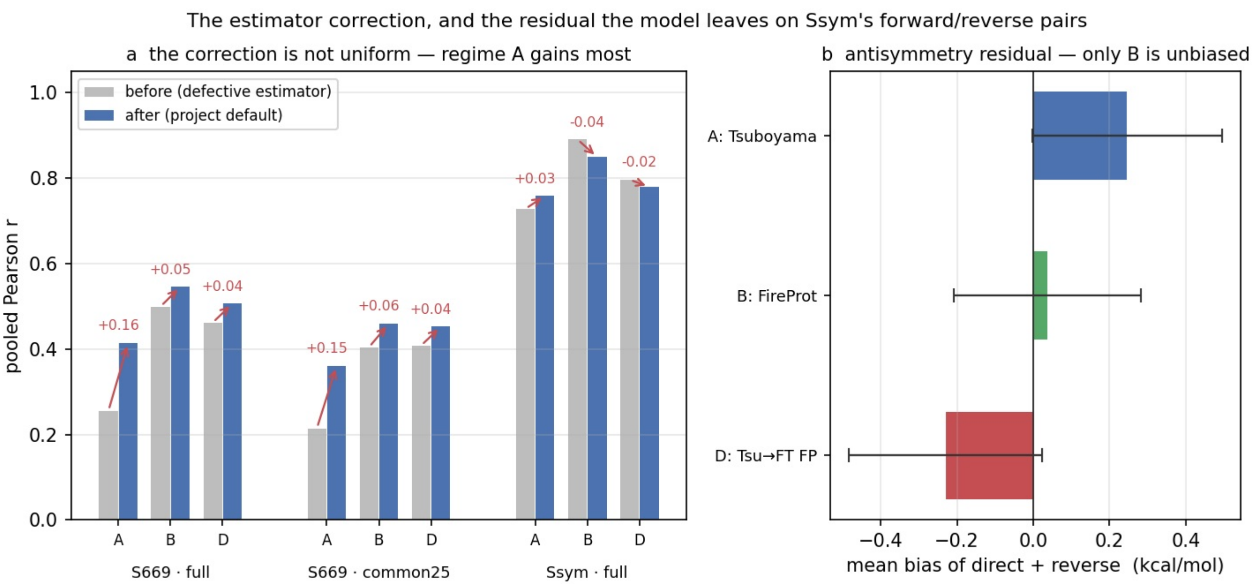

Figure 2. (a) The correction, per benchmark, subset and regime; the gain tracks training-set size rather than being uniform. (b) The residual antisymmetry bias on Ssym's forward/reverse pairs, with its standard deviation.

4. Interpretation

Three claims survive both the homology filter and the estimator correction.

Leakage is large enough to invert a conclusion, and it is measurable. Ssym's apparent ranking of training corpora disappears entirely once shared folds are removed. Any benchmark comparison that does not report an identity-controlled subset should be treated as uninformative about generalisation.

The two benchmarks are not interchangeable. Ssym is narrow and easy; every regime scores between 0.76 and 0.85 on it. S669 is diverse and hard, and it is the one that discriminates.

Within-protein ranking and cross-protein calibration are different capabilities, and this representation has much more of the first. On clean S669 the per-protein median is roughly 0.59 while pooled r is roughly 0.36. The defensible claim for this predictor is that it ranks mutations within a fold; the cross-protein scale is where the error lives.

5. Limitations

- The common-clean Ssym subset holds only 47 variants over a handful of proteins. It is unbiased but weak, and the near-equality of the three regimes there should not be read as a precise equivalence.
- Both benchmarks are capped at 500 residues, so the absolute values are not directly comparable to published figures computed on the complete sets, and no claim of state-of-the-art performance is made.
- Homology is controlled by sequence identity. Two proteins below 25 % identity can still share a fold, so the filtered subsets bound sequence-level leakage, not structural similarity.
- Ssym's reverse direction is obtained from the model's own antisymmetry rather than from an independent prediction, so §3.3 measures the internal consistency of the representation, not agreement with an external reverse-mutation calculation.

6. Conclusion

Asked how a frozen-trunk embedding ΔΔG predictor behaves on the field's standard blind benchmarks, the answer depends almost entirely on whether homology is controlled and whether the estimator is sound. With both handled, the picture is consistent: strong within-protein ranking, weak cross-protein calibration, a narrow benchmark that cannot discriminate between training regimes, and a diverse one on which the training-corpus effect is real but half the size it first appeared.